DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 14: The One-Period Binomial Model & Replicating Portfolios

Building a Derivative Out of a Stock and a Bond

September 2, 2026

Today's Agenda

1. The one-period binomial model

Today's Agenda

1. The one-period binomial model
2. Building a replicating portfolio

Today's Agenda

1. The one-period binomial model
2. Building a replicating portfolio
3. A fully worked example

Today's Agenda

1. The one-period binomial model
2. Building a replicating portfolio
3. A fully worked example
4. Interpreting Delta

Learning Objectives

- Set up the one-period binomial model.

Learning Objectives

- Set up the one-period binomial model.
- Construct a replicating portfolio matching a derivative's payoff exactly.

Learning Objectives

- Set up the one-period binomial model.
- Construct a replicating portfolio matching a derivative's payoff exactly.
- Price the derivative from the replicating portfolio's cost.

Learning Objectives

- Set up the one-period binomial model.
- Construct a replicating portfolio matching a derivative's payoff exactly.
- Price the derivative from the replicating portfolio's cost.
- Interpret the hedge ratio Δ.

A Coin Flip with Money Attached

The Simplest Possible Uncertain Stock

One period: stock goes to S_u (up) or S_d (down).

A Coin Flip with Money Attached

The Simplest Possible Uncertain Stock

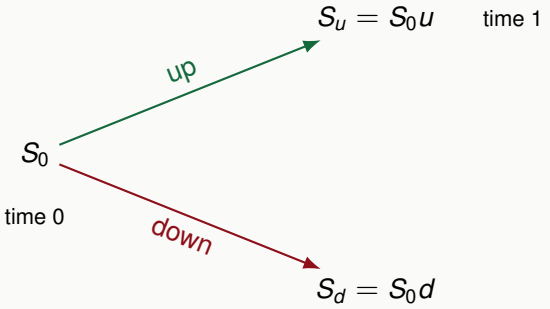
One period: stock goes to S_u (up) or S_d (down). A deliberately extreme simplification – but every idea in option pricing already appears here, uncluttered.

A Coin Flip with Money Attached

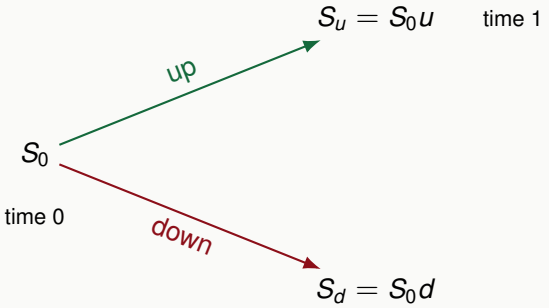
The Simplest Possible Uncertain Stock

One period: stock goes to S_u (up) or S_d (down). A deliberately extreme simplification – but every idea in option pricing already appears here, uncluttered. Master this, and multi-period (Week 6) and continuous time (Weeks 7-9) are bookkeeping, not new ideas.

The Tree



The Tree



Derivative payoffs: $C_u = \max(S_u - K, 0)$, $C_d = \max(S_d - K, 0)$ – both known.
Unknown: C_0.

Matching Payoffs Exactly

Set Up Two Equations

$$\Delta S_u + (1+r)B = C_u \qquad \Delta S_d + (1+r)B = C_d$$

Matching Payoffs Exactly

Set Up Two Equations

$$\Delta S_u + (1+r)B = C_u \qquad \Delta S_d + (1+r)B = C_d$$

Subtract to eliminate B:

$$\Delta = \frac{C_u - C_d}{S_u - S_d}$$

Matching Payoffs Exactly

Set Up Two Equations

$$\Delta S_u + (1+r)B = C_u \qquad \Delta S_d + (1+r)B = C_d$$

Subtract to eliminate B:

$$\Delta = \frac{C_u - C_d}{S_u - S_d}$$

Substitute back:

$$B = \frac{C_d - \Delta S_d}{1+r}$$

Pricing by Replication

Key Takeaway

$$C_0 = \Delta S_0 + B$$

Pricing by Replication

Key Takeaway

$$C_0 = \Delta S_0 + B$$

Same payoff in every state $\Rightarrow$ same price today (Law of One Price).

Building a Copy, Reading Its Price Tag

The Missing Toy

Can’t find the exact toy? Notice it behaves like two other toys taped together.

Building a Copy, Reading Its Price Tag

The Missing Toy

Can't find the exact toy? Notice it behaves like two other toys taped together. Buy those, tape them – now you own something behaving identically.

Building a Copy, Reading Its Price Tag

The Missing Toy

Can't find the exact toy? Notice it behaves like two other toys taped together. Buy those, tape them – now you own something behaving identically. Whatever you paid for the combo is what the missing toy should cost too.

Setting Up the Numbers

S_0=100, u=1.2, d=0.9, r=5%, K=105

$$C_u = \max(120 - 105, 0) = 15 \qquad C_d = \max(90 - 105, 0) = 0$$

Solving for Δ and B

$$\Delta = \frac{15 - 0}{120 - 90} = 0.5$$

Solving for Δ and B

$$\Delta = \frac{15 - 0}{120 - 90} = 0.5$$

$$B = \frac{0 - 0.5(90)}{1.05} = \frac{-45}{1.05} \approx -42.857$$

Solving for Δ and B

$$\Delta = \frac{15 - 0}{120 - 90} = 0.5$$

$$B = \frac{0 - 0.5(90)}{1.05} = \frac{-45}{1.05} \approx -42.857$$

(Negative B: borrowing Rs. 42.857 at the risk-free rate.)

The Price, and the Check

$$C_0 = 0.5(100) - 42.857 \approx \mathbf{7.143}$$

The Price, and the Check

$$C_0 = 0.5(100) - 42.857 \approx \mathbf{7.143}$$

	Replica value	Derivative payoff
Up	$60 - 45 = 15$	15 ✓
Down	$45 - 45 = 0$	0 ✓

Delta: The Hedge Ratio

Definition

$$\Delta = \frac{C_u - C_d}{S_u - S_d}$$

shares of stock needed, per derivative, to replicate/hedge its payoff.

Delta: The Hedge Ratio

Definition

$$\Delta = \frac{C_u - C_d}{S_u - S_d}$$

shares of stock needed, per derivative, to replicate/hedge its payoff.

A first glimpse of the Greeks

Δ is literally the price's sensitivity to S_0 – the same "Delta" you'll meet in continuous time, Week 10.

Notice What We Never Used

Key Takeaway

No real-world probability of the stock going up. No investor's risk aversion. No market equilibrium assumption. $C_0 = 7.143$ follows purely from S_0, u, d, r, K.

Key Takeaways

Key Takeaway

- Binomial model: $S_0 \to S_u$ or S_d; bond grows at r.

Key Takeaways

Key Takeaway

- Binomial model: $S_0 \to S_u$ or S_d; bond grows at r.
- Replicating portfolio matches payoff in both states.

Key Takeaways

Key Takeaway

- Binomial model: $S_0 \to S_u$ or S_d; bond grows at r.
- Replicating portfolio matches payoff in both states.
- $C_0 = \Delta S_0 + B$ by the Law of One Price.

Key Takeaways

Key Takeaway

- Binomial model: $S_0 \to S_u$ or S_d; bond grows at r.
- Replicating portfolio matches payoff in both states.
- $C_0 = \Delta S_0 + B$ by the Law of One Price.
- $\Delta = (C_u - C_d)/(S_u - S_d)$ – hedge ratio and first Greek.

Key Takeaways

Key Takeaway

- Binomial model: $S_0 \to S_u$ or S_d; bond grows at r.
- Replicating portfolio matches payoff in both states.
- $C_0 = \Delta S_0 + B$ by the Law of One Price.
- $\Delta = (C_u - C_d)/(S_u - S_d)$ – hedge ratio and first Greek.
- Price never used p or risk preferences.

Check Your Understanding

Discuss (2 minutes)

1. S_0=50, u=1.1, d=0.85, r=4%, put K=52. Find C_u, C_d, Δ, B, C_0.

Check Your Understanding

Discuss (2 minutes)

1. S_0=50, u=1.1, d=0.85, r=4%, put K=52. Find C_u, C_d, Δ, B, C_0.
2. Why must a call's Δ lie in $[0, 1]$?

Check Your Understanding

Discuss (2 minutes)

1. $S_0 = 50$, $u = 1.1$, $d = 0.85$, $r = 4\%$, put $K = 52$. Find C_u, C_d, Δ, B, C_0.
2. Why must a call's Δ lie in $[0, 1]$?
3. Someone offers the worked-example call for Rs. 6.50. What's the arbitrage?

Looking Ahead

Solving two equations every time is tedious.

Looking Ahead

Solving two equations every time is tedious.

Risk-Neutral Probabilities – Lecture 15